

Hamse Abdi

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TECHNICAL SKILLS

AI & Agents	Multi-agent system design · context engineering · RAG & hybrid search (OpenSearch) · agent memory (short/long-term) · LLM-as-judge & trajectory evals · observability/tracing (Langfuse, OTEL) · guardrails & prompt-injection defense · prompt engineering (Pydantic structured outputs, caching) · MCP tool use
Models & Platforms	AWS Bedrock · Bedrock AgentCore · Strands Agents SDK · Claude (API) · Amazon Nova · Azure OpenAI
Infrastructure & DevOps	AWS Dev Certificate (DVA-C02) · Lambda · EKS · Docker · DynamoDB · MongoDB · SQS · RabbitMQ · EventBridge · Terraform · Jenkins · GitHub Actions
Code & Architecture	Python · TypeScript / JS · Node · React · Next · Flask · REST / OpenAPI design · event-driven, microservice & serverless architecture · data modeling · CI / CD · OIDC · OAuth 2.0 / Okta · Snyk · pytest · moto

EXPERIENCE

Software Engineer — AI Systems

June 2021 – Present

Travelers · Atlanta, GA

- **Built and shipped a generative-AI underwriting workbench that cut case review from roughly 6 hours to 30 minutes**, freeing a capacity-constrained underwriting team to clear far more volume. It summarizes hundreds of claim notes and unifies policy data, claim history, photos, and geolocation into a single decision view.
- **Evolved it into an agentic system** — a reader agent and a panel of parallel, rubric-driven evaluator agents — that automatically renews clear-cut policies and escalates the rest with cited, reason-coded referrals, keeping an underwriter in the loop on every non-renewal.
- **Designed the end-to-end eval pipeline for the agentic underwriting system** — a ground-truth labeling workflow run with underwriter SMEs, rubrics that turn their judgment into scorable criteria, and an LLM-as-judge that grades agent outputs against those rubrics for SME review — enabling rapid, data-driven prompt tuning.
- **Built the agentic pipeline and the tools that power it** — MCP tools for on-the-fly data retrieval, an AWS Step Functions + Lambda orchestrator that preloads context and refreshes large, complex policy PDFs into structured, model-readable records in MongoDB, and externalized prompts, skills, and output rules in S3 — so the Agents behavior can be tuned through configuration instead of code changes.
- **Drove the system from prototype to production** with Legal, Compliance, AI Risk, and the architecture team — co-designing a scalable runtime on **Bedrock AgentCore** (serverless agents, MCP tool gateway, Okta, Terraform); config-driven, so new evaluators ship without code changes.
- **Led a small team to own a RESTful API platform end-to-end** — OpenAPI contracts, Okta OAuth 2.0 SSO (Lambda Authorizer, Lambda@Edge), event-driven services (Lambda, SQS, EventBridge) on EKS, Terraform IaC, and the CI/CD pipeline — adopted by **three product teams** and the infrastructure today's AI systems are built on.
- **Set technical direction and scaled the team** — authored design docs and RFCs, presented AI systems to senior leadership, and ran internal generative-AI training to upskill the broader engineering org; mentored **3 engineers and 2 interns**; served as **Security Champion**, embedding Snyk scanning across **5 teams**.

PERSONAL PROJECTS

Stride — Proactive AI Productivity Coach *Strands SDK · Claude Sonnet 4.6 · DynamoDB · Twilio · Lambda · Terraform*

- **Texts users first** — no app, no login, no dashboard. Claude orchestrates 19 tools across the full coaching loop: goal-setting, daily progress, weekly retrospectives, and mid-week adjustments — all in plain English.
- **Builds a behavioral profile per user** — commit accuracy, pacing trends, and recurring blockers — so coaching adapts to each person over time. Two-model cost design: Haiku classifies intent at ~\$0.001/call; Sonnet handles reasoning with prompt caching.
- **Shipped with real guardrails** — 233-test suite (fast CI gate <5 s + nightly Opus LLM-as-judge, critique-then-verdict); TCPA opt-out enforcement; per-user timezone scheduling via EventBridge.

Stock Intel — Telegram Market Intelligence Platform *Python · Claude API · Lambda (ARM64) · DynamoDB · SQS · EventBridge · Terraform*

- **Serverless AWS platform**: 27 commands (portfolio P&L, options Greeks, dark pool, congressional trades, technical signals), 8 EventBridge-scheduled scanners pushing SQS-delivered Telegram alerts — 286 tests, ~\$2/month.
- **Cost-efficient LLM architecture**: deterministic signal scoring across 5 data sources (yfinance, FINRA, Capitol Trades, Yahoo RSS) feeds pre-computed summaries to Claude — 80% of commands cost \$0, AI synthesis calls ~\$0.01 each.
- **Production eval loop**: conviction-gated recommendations are persisted at entry price and auto-graded against actual returns at 7, 14, and 30-day horizons.

Apex — Protocol-Driven Health Accountability Agent *Strands SDK · Bedrock · Telegram Bot API · EventBridge · DynamoDB · Lambda*

- **Self-configuring agent architecture** — add a metric to apex.yaml and the agent gains typed log / get tools at next cold start via Python closures over the Strands SDK; zero code changes required; 25 moto integration tests validate every tool.
- **Natural-language Telegram interface** — “slept 7h, 180g protein” → Claude parses and routes to the right tool automatically; single-table DynamoDB + GSI for per-metric history; EventBridge-scheduled morning check-ins keep the user accountable.

EDUCATION

University of Minnesota – Twin Cities

College of Science and Engineering — B.S. Computer Science

May 2021

GPA: 3.7 / 4.0